

Desigualdad de Hölder

Teorema 1 (Desigualdad de Hölder). Sea $p \in (1, \infty)$ y $f \in \mathcal{L}^p(\mu)$, $g \in \mathcal{L}^q(\mu)$

$$\implies \|f \cdot g\|_1 \leq \|f\|_p \cdot \|g\|_q$$

donde q es el exponente conjugado de p .

Demostración: Si $\|f\|_p = 0$ o $\|g\|_q = 0$, la desigualdad es trivial, por tanto, suponemos que $\|f\|_p \neq 0$ y $\|g\|_q \neq 0$ y definimos $\widehat{f} := f/\|f\|_p$ y $\widehat{g} := g/\|g\|_q$.

$$\implies \|f \cdot g\|_1 = \int_{\Omega} |f(x) \cdot g(x)| \, d\mu(x) = \|f\|_p \cdot \|g\|_q \int_{\Omega} |\widehat{f}(x) \cdot \widehat{g}(x)| \, d\mu(x).$$

Por la desigualdad de Young, $\forall x \in \Omega : \left| \widehat{f}(x) \cdot \widehat{g}(x) \right| \leq \frac{|\widehat{f}(x)|^p}{p} + \frac{|\widehat{g}(x)|^q}{q}$.

$$\begin{aligned} \implies \|f \cdot g\|_1 &\leq \|f\|_p \cdot \|g\|_q \left(\frac{1}{p} \int_{\Omega} |\widehat{f}(x)|^p \, d\mu(x) + \frac{1}{q} \int_{\Omega} |\widehat{g}(x)|^q \, d\mu(x) \right) = \\ &\leq \|f\|_p \cdot \|g\|_q \left(\frac{1}{p} + \frac{1}{q} \right) = \|f\|_p \cdot \|g\|_q. \end{aligned}$$

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